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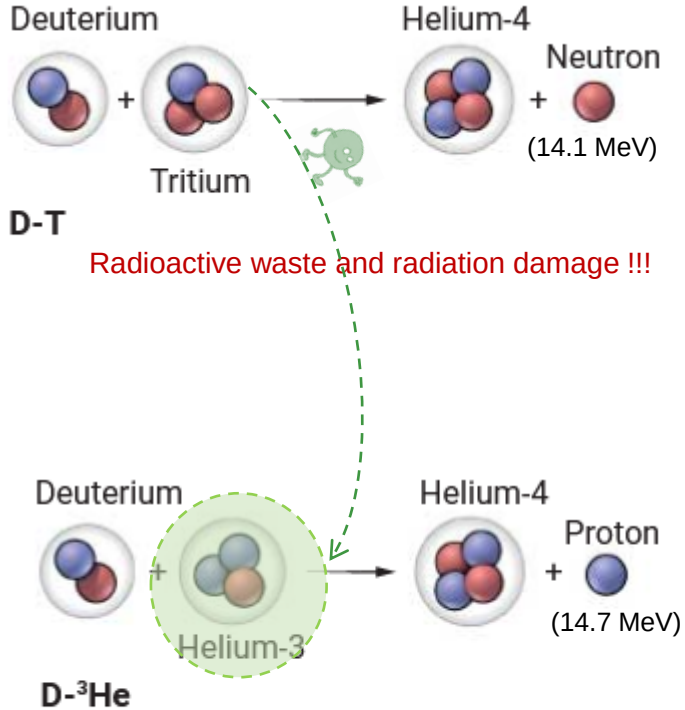
Research Training Group 2721: HYDROGEN Isotopes $^1,^2,^3\text{H}$

Rational design of materials for adsorptive He isotope separation

Elvira Gouatieu Dongmo

Supervision: Prof. Dr. Thomas Heine
Retreat - Dresden, 30.09.2024

Why helium?



New Way to Achieve Nuclear Fusion!!!



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Fundamental Plasma Physics

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A D-³He fusion reactor for the mitigation of global warming

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Retired from Princeton University Plasma Physics Laboratory, 70 Riverside Drive, Princeton, NJ 08540, USA



ARTICLE INFO

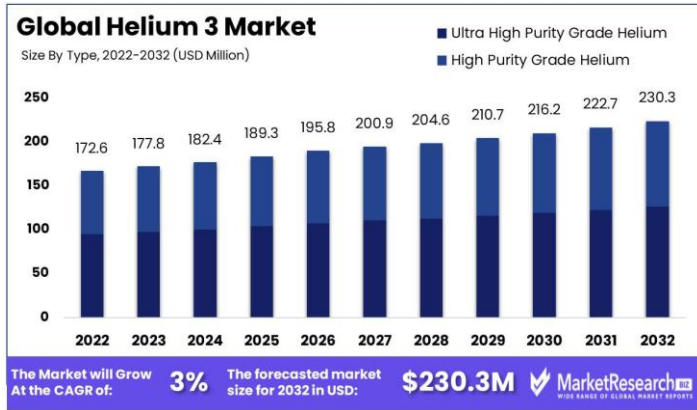
Keywords:

Helium-3
Racetrack configuration
Helical windings
Magnetic shear
Bremsstrahlung

ABSTRACT

Since a fusion reactor using the **Deuterium-Tritium fuel cycle cannot be a source of clean energy** because of the deleterious effects of energetic neutrons carrying 80% of the energy output, and it is very doubtful that it will be able to achieve Tritium self-sufficiency because of an extremely problematic and still unproven breeding procedure, **this paper proposes a new reactor scheme capable of confining hot and dense plasmas using the Deuterium – Helium-3 fuel cycle.** Such a reactor must be considered a source of clean energy because of its very low level of neutrons production, and its fuel is available in large quantity since we can get the needed Deuterium from seawater and likewise Helium-3 from the moon, as it was found from the samples of lunar soil brought back by the astronauts of the Apollo Mission. The proposed reactor consists of two 100 m long cylindrical plasmas,

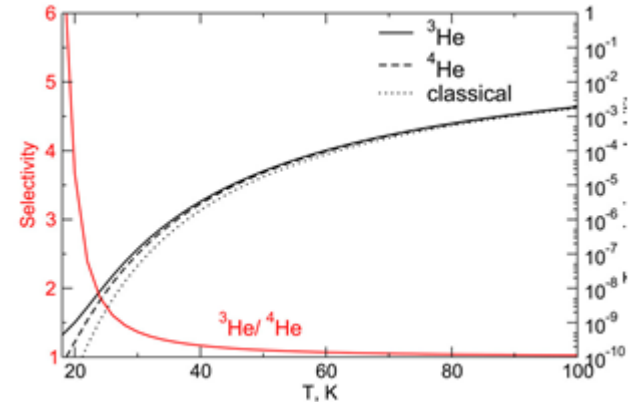
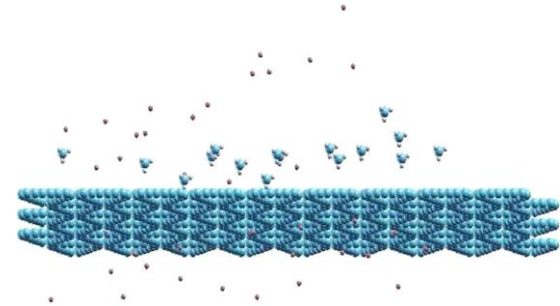
Strategies for obtaining Helium-3



Helium-3 (^3He) is:

- an isotope of helium, rare on Earth and hard to acquire;
- an indispensable resource for scientific research;
- has applications in a variety of industries;
- must be produced using efficient methods.**

Two-dimensional (2D) materials:
He through graphdiyne



$^3\text{He}/^4\text{He}$ selectivity of 6 at about 20 K

[3] Source: marketresearch.biz/report/helium-3-market/, (2023)

[4] Bartolomei, Massimiliano, et al. *The Journal of Physical Chemistry C* 118.51 (2014): 29966-29972.

Alternative strategies for obtaining helium-3

Separation techniques

Cryogenic distillation

Pressure swing adsorption (PSA)

Membrane separation from natural gas

Very expensive

Low temperature

Quantum sieving approaches

Literature on isotope-selective He adsorption



[5] Dai, Zhongde, et al. Separation and Purification Technology 274 (2021): 119044.

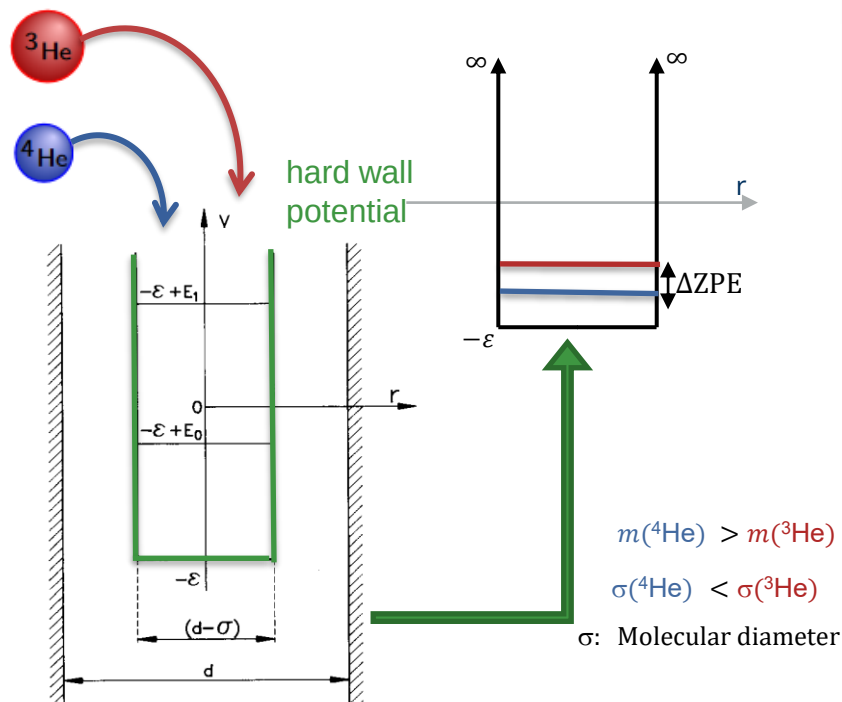
[6] T.E. Rufford, K.I. Chan, S.H. Huang, E.F. May, Adsorpt. Sci. Technol. 32 (2014) 49–72.

[7] Quet al.Sci. Rep. (2016), 19952. doi:10.1038/srep19952

KQS and CAQS principles

KQS

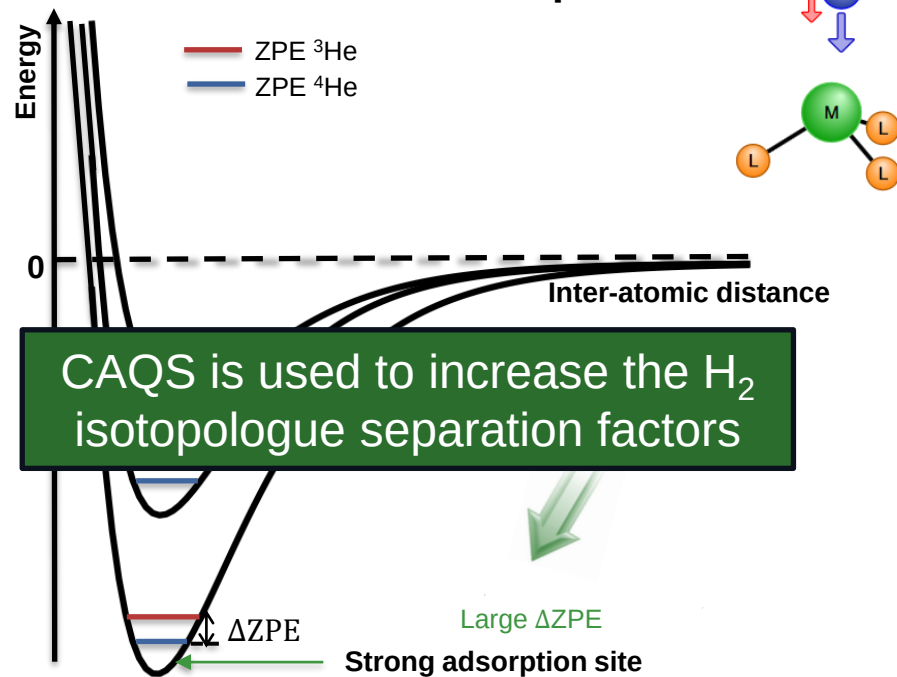
Selective pore diffusion



KQS = Kinetic Quantum Sieving

CAQS

attractive surface adsorption site

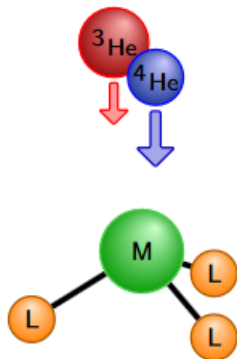


CAQS is used to increase the H_2 isotopologue separation factors

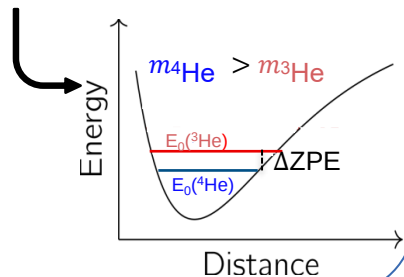
CAQS = Chemical Affinity Quantum Sieving

Research aim

Chemical Affinity Quantum Sieving (CAQS)

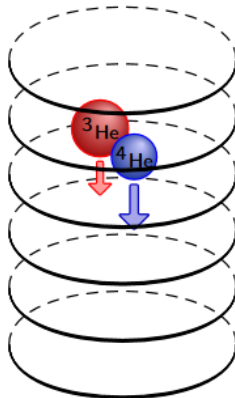


attractive surface adsorption site



+

Kinetic Quantum Sieving (KQS)



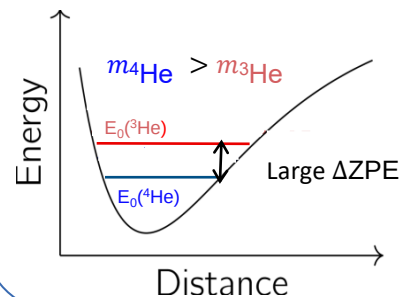
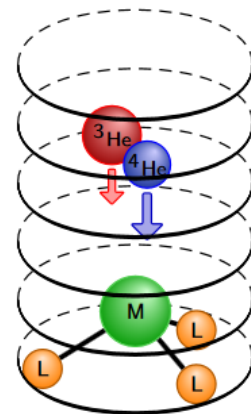
selective pore diffusion

$$E_0 \sim \frac{1}{m(d - \sigma)^2}$$

- E_0 : Zero-point energy
- d : Pore diameter
- σ : Molecular diameter:
- $(d - \sigma) \sim$ De Broglie wavelength.

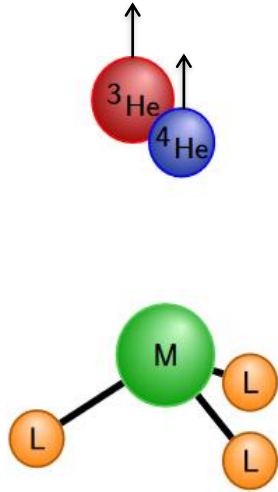


CAQS + KQS

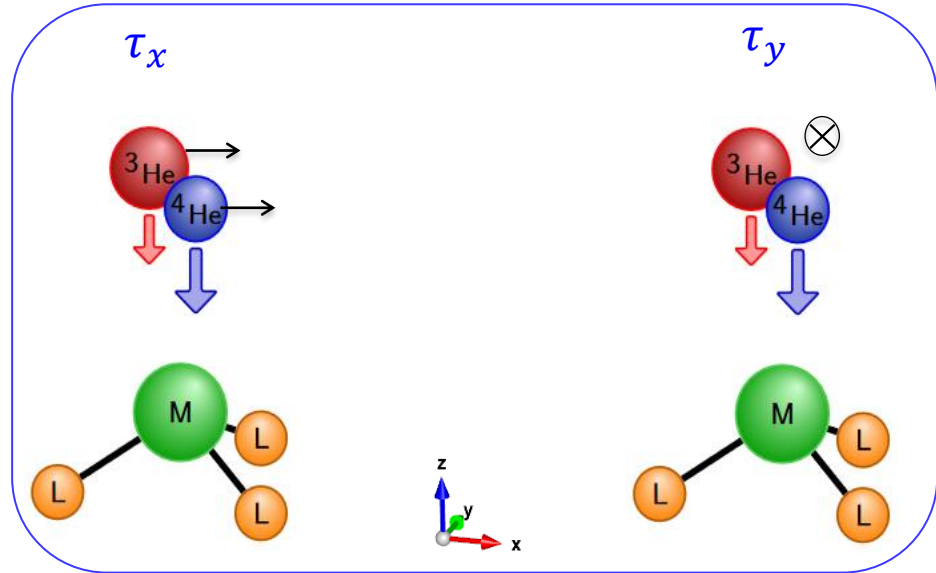


Mechanism of selective adsorption

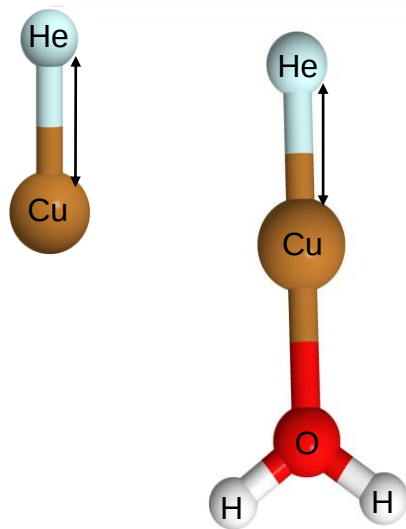
03 degrees of freedom



vibration



translations

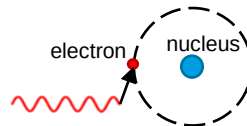


Copper(I)–helium interaction
– selected system for efficient adsorptive He isotope separation via numerical methods

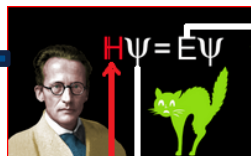
Computational approaches

☞ Electron much lighter than nuclei, thus much faster : $V_N \ll V_e$

☞ The Born-Oppenheimer approximation



Schrödinger Equation



Energy "Eigenvalue"

Wavefunction

Hamiltonian Operator

Electronic SE

Nuclear SE

**ab initio
calculations**

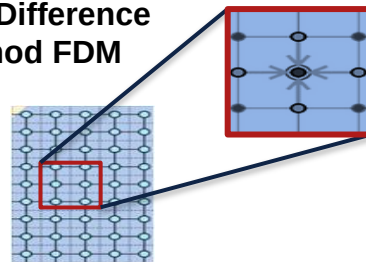
Numerical methods

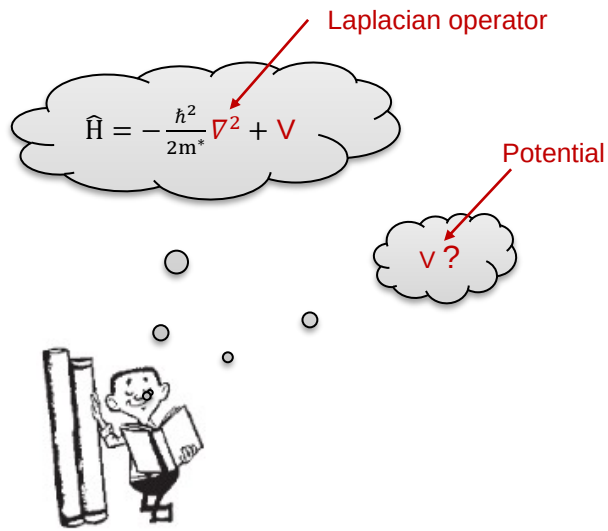
(CCSD(T) / aug-cc-pVTZ)

Geometry
Electronic energy

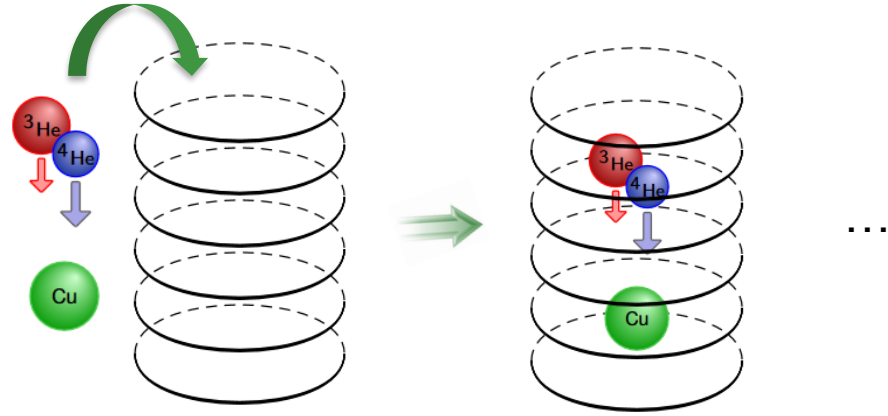
**Potential energy
surface (PES)**

**Finite Difference
Method FDM**





m^* : reduced mass; $\hbar$: reduced Planck constant



Numerical solutions

Workflow process overview: potential profiles

r : position in the (x, y) plane

r_0 : initial radius of the wall

V_{conf} : confining pore potential

$$V(r, z) = V_{\text{Conf}}(r) + V_{\text{Morse}}(z)$$

$$|r| < r_0$$

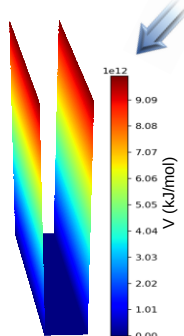
true ($V_{\text{Conf}} = 0$)

false ($V_{\text{Conf}} \neq 0$)

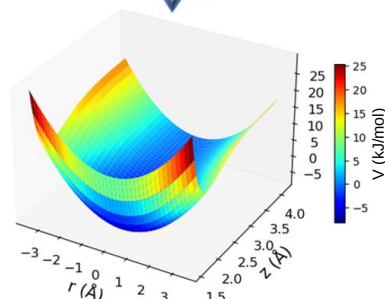
hard wall
model

soft wall models

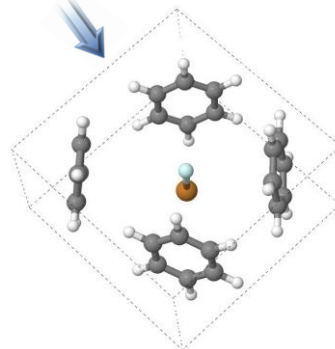
Morse
potential



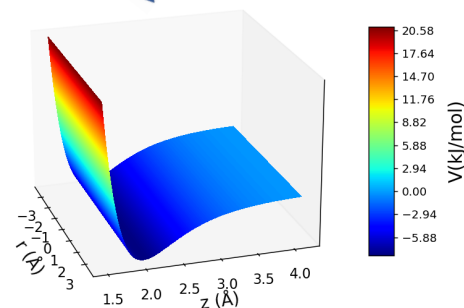
Infinite potential



harmonic potential

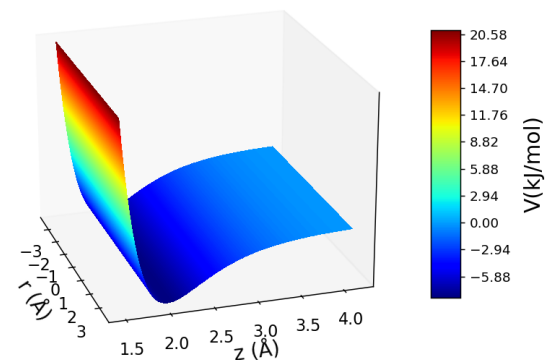


Molecular confinement



Nuclear Quantum Effects (NQE)

Morse potential

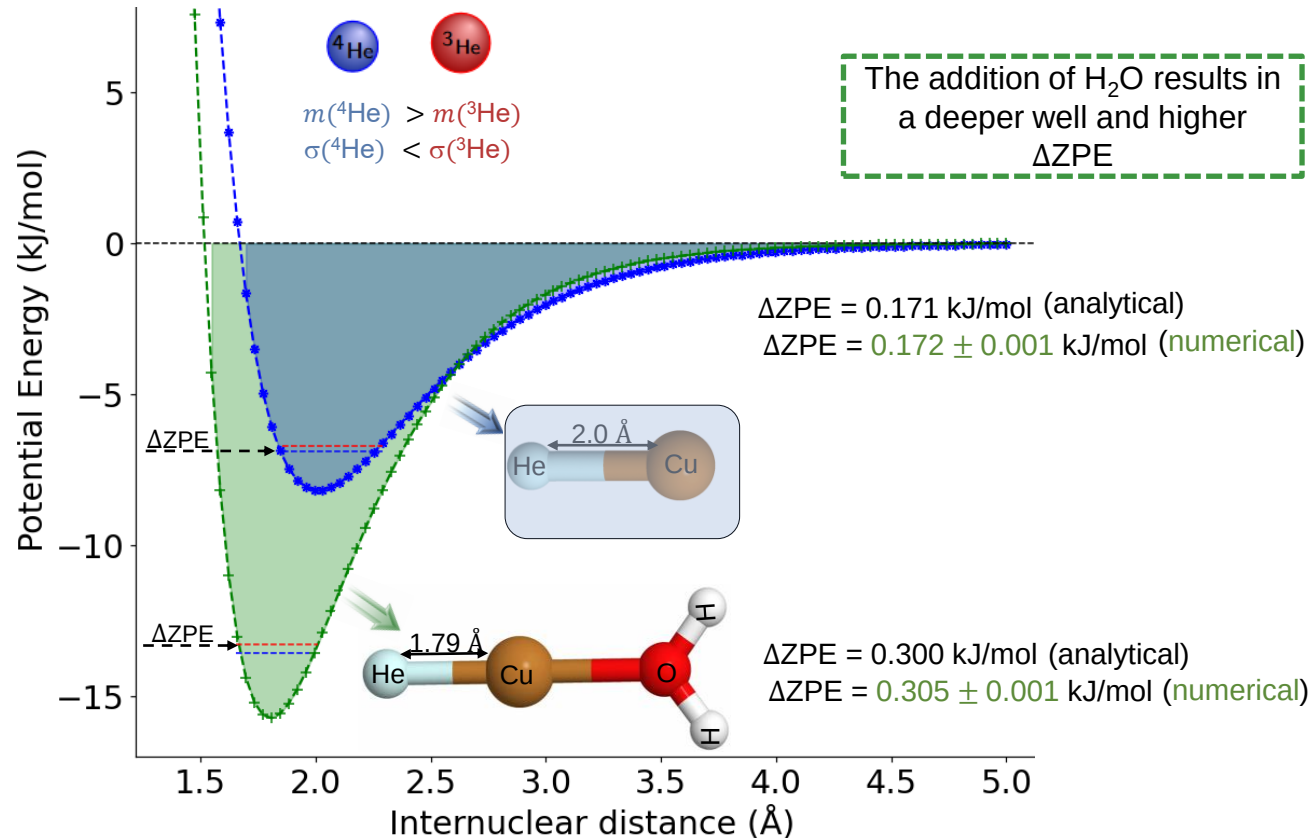


$N = 66 \times 66 \times 66$ points;

grid spacing: $\begin{cases} \Delta r \text{ (step)} = 0.09 \text{ \AA} \\ \Delta z \text{ (step)} = 0.03 \text{ \AA} \end{cases}$

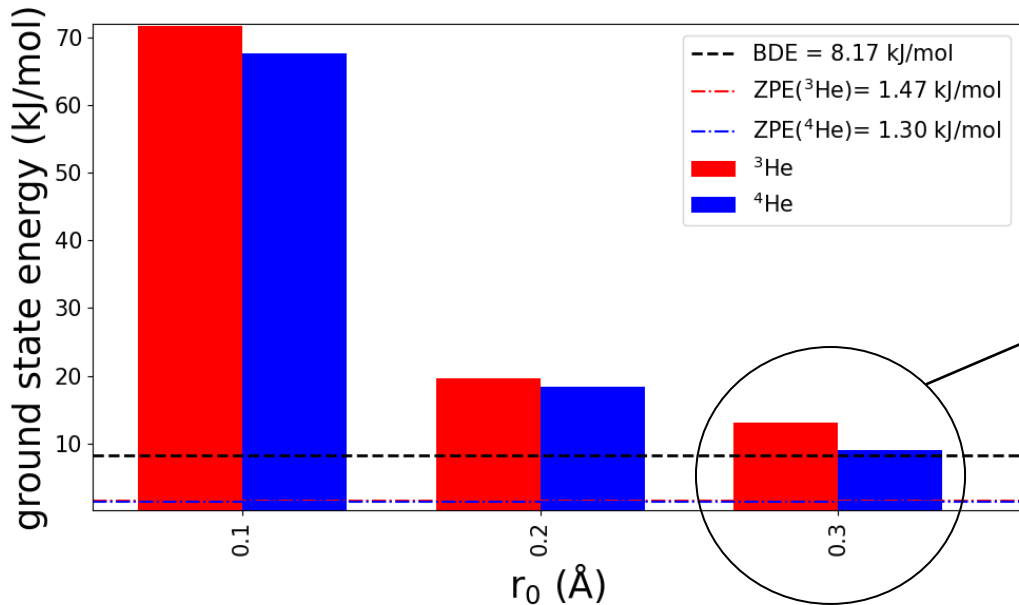
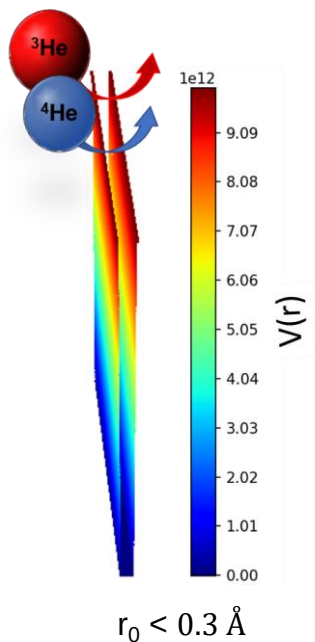
0.001 is the truncation error

Optimized geometry:
CCSD(T)/aug-cc-pVTZ-PP level

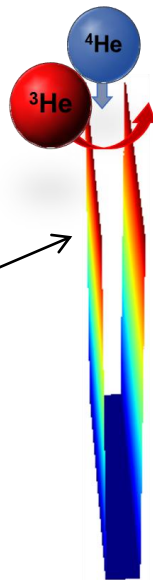


Confining pore potentials: Hard wall model

The pore radius is very small:
no physical penetration

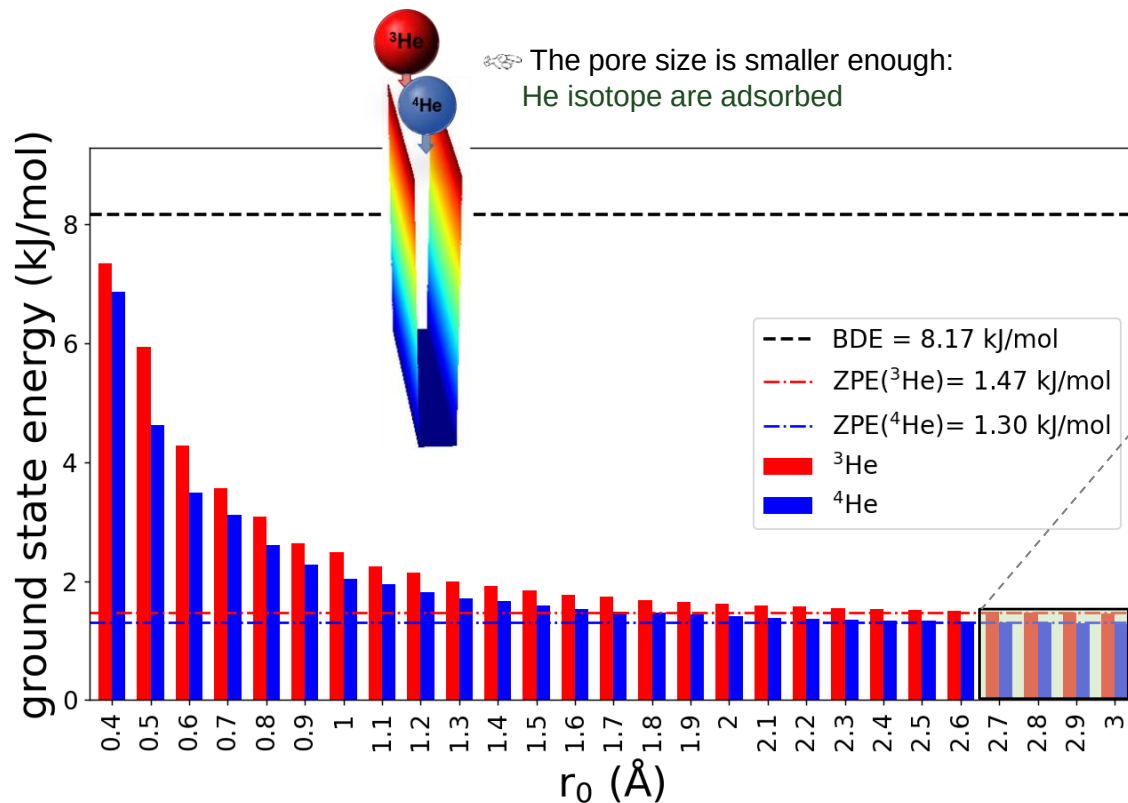


molecular sieving

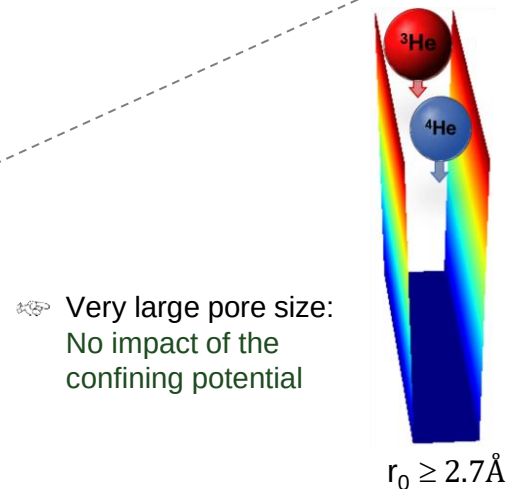
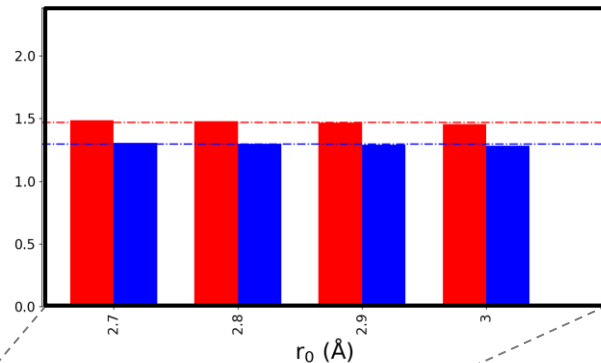


grid spacing : $\Delta r \text{ (step)} = 0.003 \text{ \AA}$; $\Delta z \text{ (step)} = 0.001 \text{ \AA}$

Confining pore potentials: hard wall model



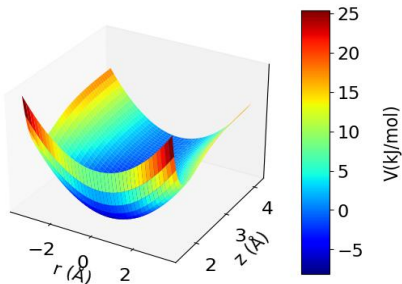
grid spacing : Δr (step) = 0.003 Å ; Δz (step) = 0.001 Å



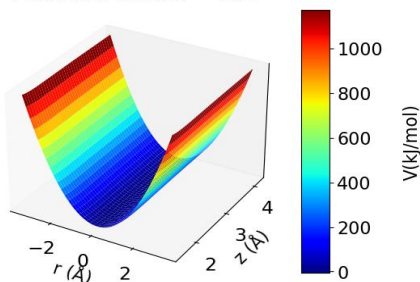
Confining pore potentials: soft wall model

$$V_{c_sw}(r) = \begin{cases} 0, & \text{if } |r| \leq r_0 \\ k(r - r_0)^2 & \text{otherwise.} \end{cases}$$

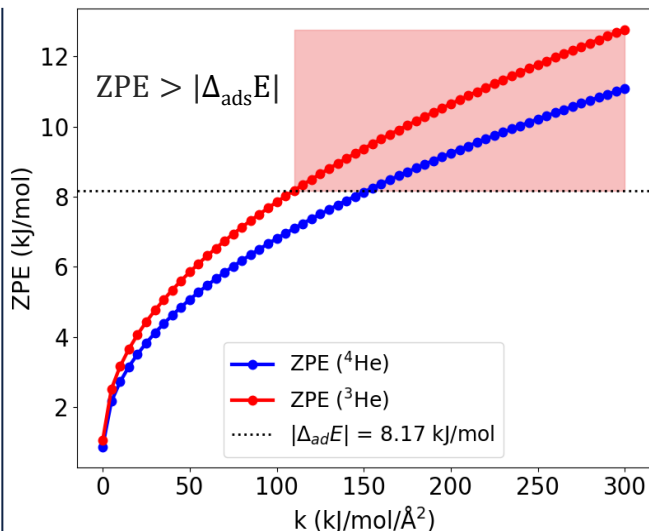
Potential with $k = 1.5$



Potential with $k = 100$

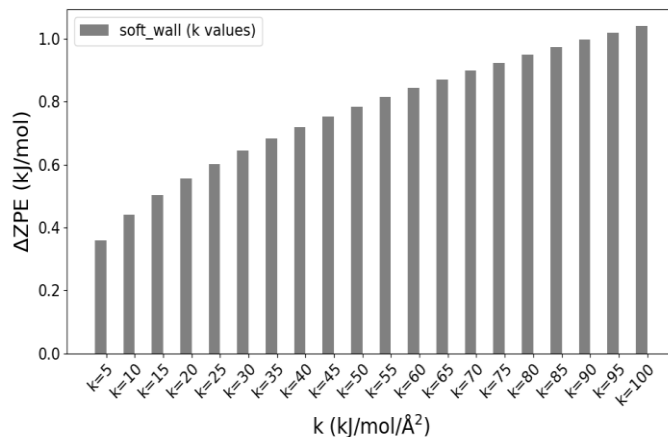


k = the spring constant

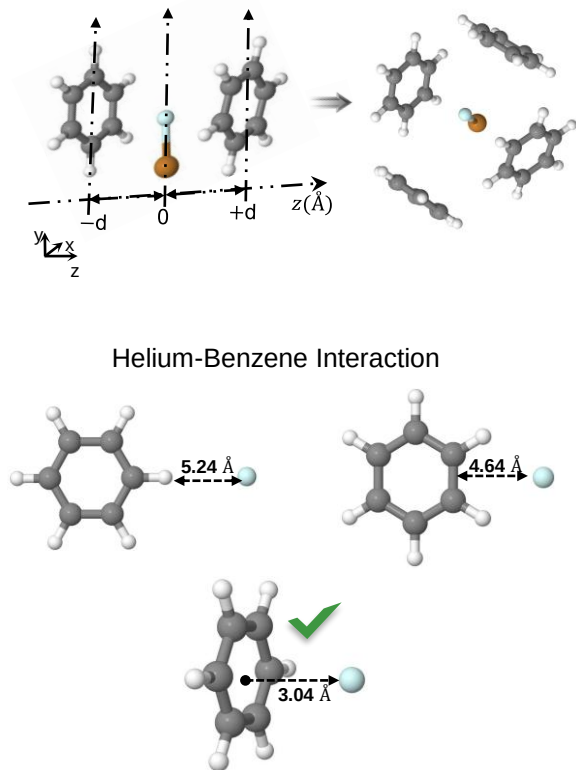


ΔZPE ($^3\text{He}/^4\text{He}$) increases with increasing k

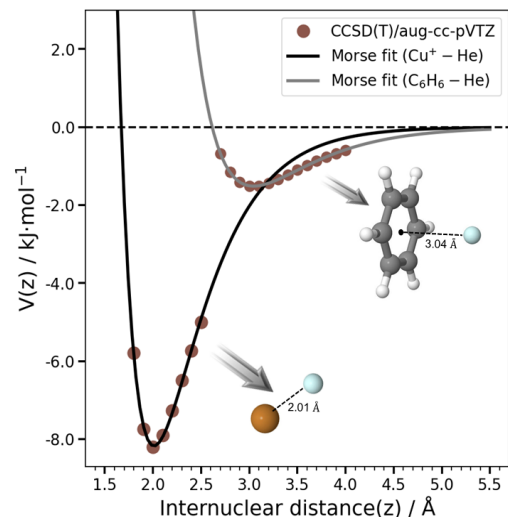
High k value leads to a significant rise in ZPE, making adsorption impossible



Molecular confinement: more realistic potential

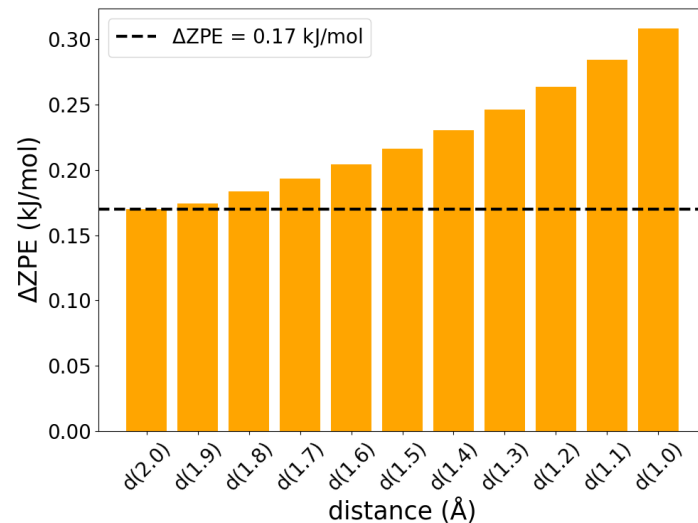


d : distance between the benzene ring and He



BDE(De) of the $\text{He}-\text{C}_6\text{H}_6$ is **1.51 kJ/mol**

BDE(De) of the $\text{He}-\text{Cu}^+$ is **8.17 kJ/mol**



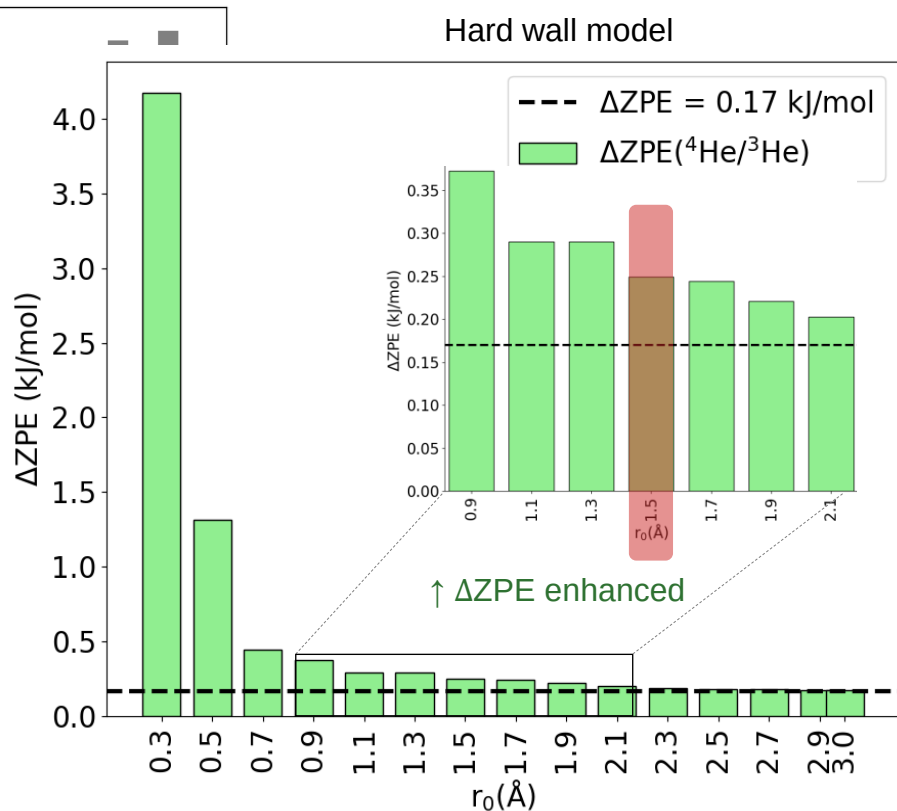
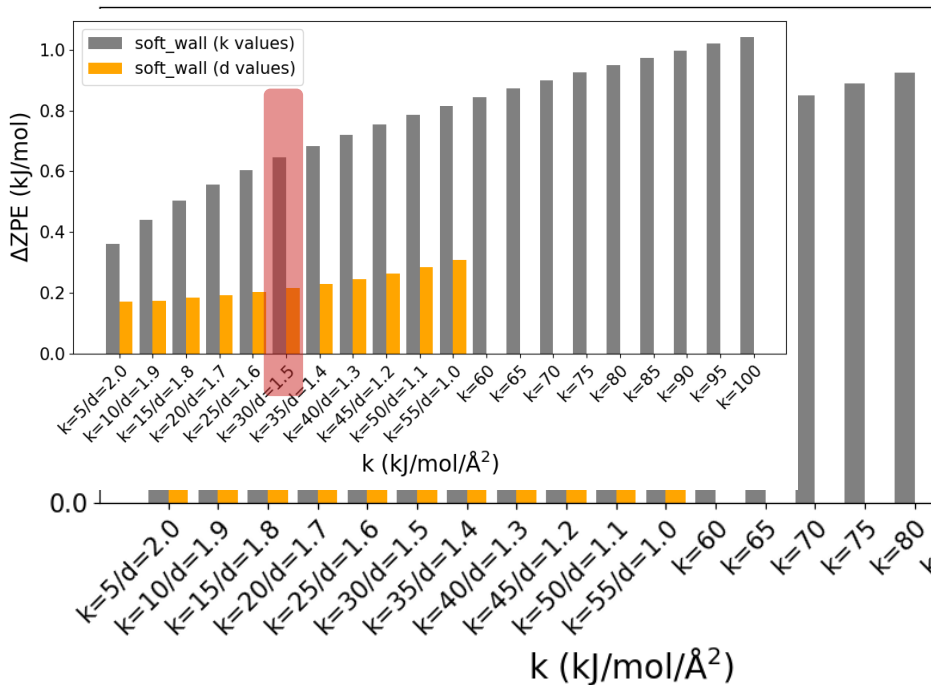
Slight increase in ΔZPE



ΔZPE increases with the decrease of the distance d

$N = 66 \times 66 \times 66$ points; grid spacing: Δz (step) = $0.1 \text{ \AA}$

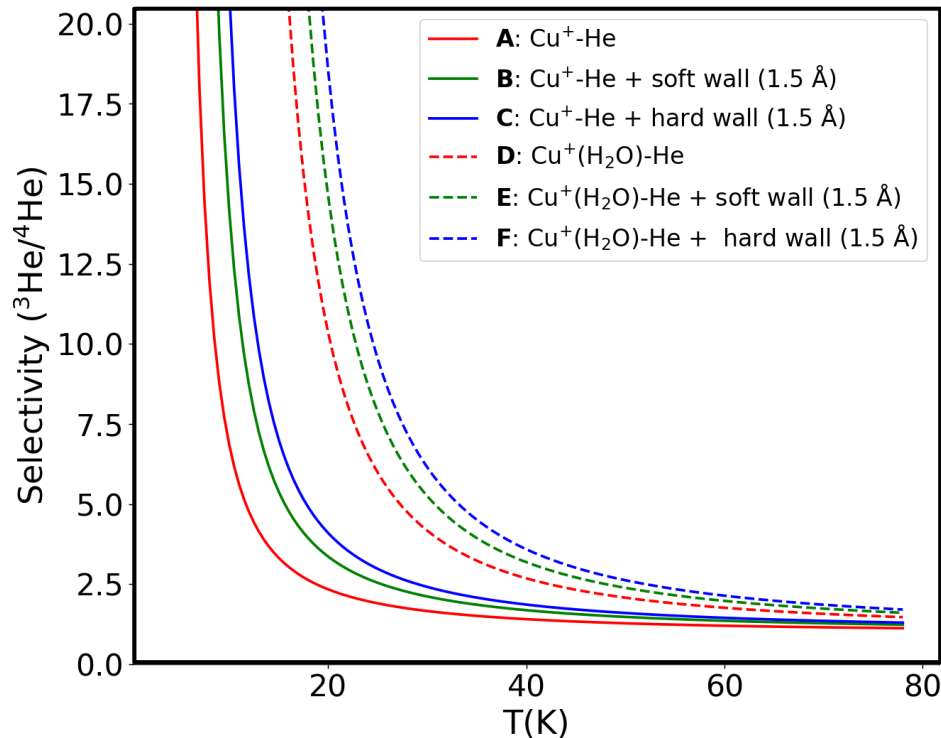
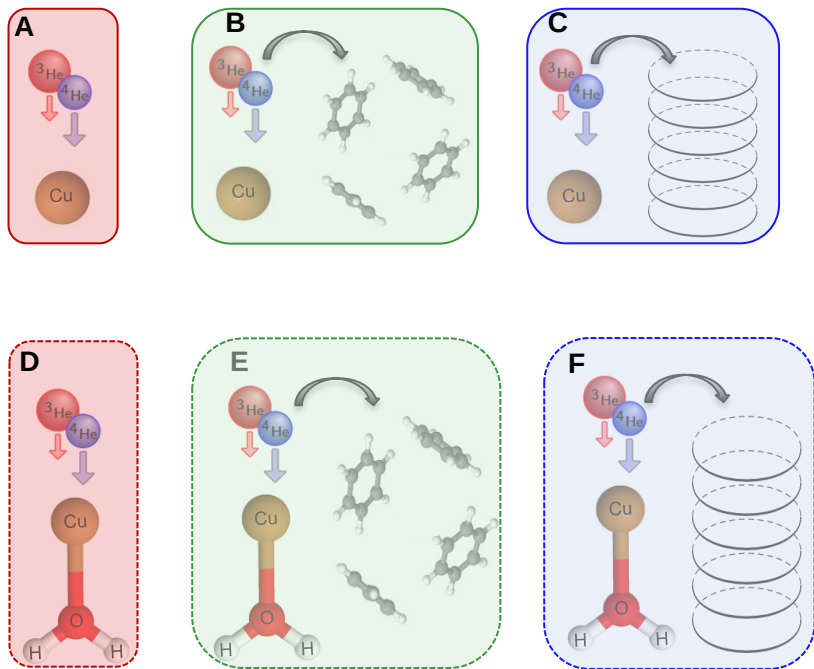
ΔZPE ($^4\text{He}/^3\text{He}$) enhanced?



🐼 The soft wall model shows a higher ΔZPE ($^3\text{He}/^4\text{He}$) with increasing k -values compare to the confinement by 4 benzene rings confinement.

$\Delta_{\text{ads}}E$ ↓ binding weakened

Predicted selectivities for He isotope at Cu^+ and $\text{Cu}^+(\text{H}_2\text{O})$



☞ Separation factors at low temperatures regime (below 77K) significantly increase in presence of the confinements

Summary

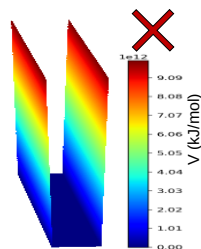
$\Delta\text{ZPE}({}^3\text{He}/{}^4\text{He})$ enhanced?

Trade-off:

$\Delta\text{ZPE}({}^3\text{He}/{}^4\text{He}) \uparrow$ ΔZPE enhanced

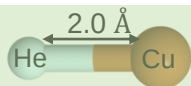
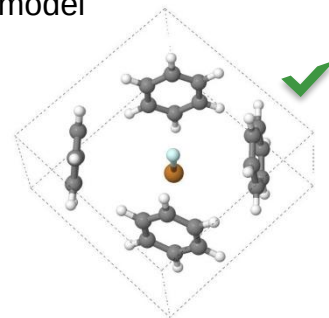
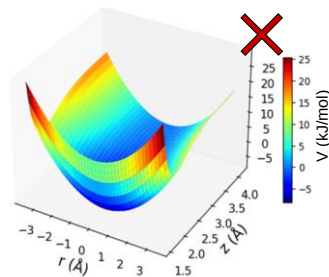
$\Delta_{\text{ads}}E \downarrow$ binding weakened

Hard wall model



$\Delta\text{ZPE}({}^3\text{He}/{}^4\text{He})$ INCREASES

Soft wall model

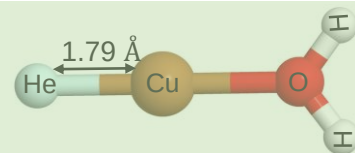


0.17 kJ/mol
without confinement

0.25 kJ/mol with
 $r_0 = 1.5 \text{ \AA}$

0.64 kJ/mol with
 $k = 30 \text{ kJ/mol/\AA}^2$

0.22 kJ/mol with
 $d = 1.5 \text{ \AA}$



0.31 kJ/mol
without confinement

0.39 kJ/mol with
 $r_0 = 1.5 \text{ \AA}$

0.66 kJ/mol with
 $k = 30 \text{ kJ/mol/\AA}^2$

0.35 kJ/mol with
 $d = 1.5 \text{ \AA}$

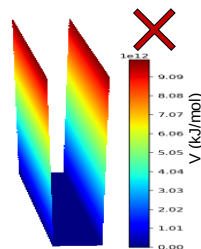
Summary: $^3\text{He}/^4\text{He}$ isotope separation

Selectivity enhanced?

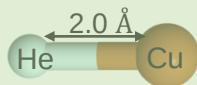
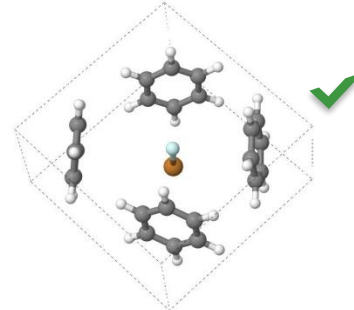
cryogenic temperatures

$^3\text{He}/^4\text{He}$ Selectivity INCREASES

Hard wall model



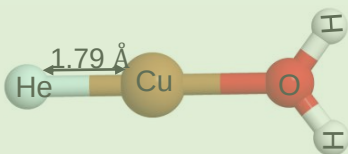
Soft wall model



$\alpha = 30.9$ at $T = 6$ K
 $\alpha = 1.5$ at $T = 38$ K

$\alpha = 201.0$ at $T = 6$ K
 $\alpha = 2.0$ at $T = 38$ K

$\alpha = 104.0$ at $T = 6$ K
 $\alpha = 1.8$ at $T = 38$ K



$\alpha = 98.7$ at $T = 8$ K
 $\alpha = 2.87$ at $T = 38$ K

$\alpha = 284.3$ at $T = 8$ K
 $\alpha = 3.90$ at $T = 38$ K

$\alpha = 184.6$ at $T = 8$ K
 $\alpha = 3.44$ at $T = 38$ K

not accurate

better prediction

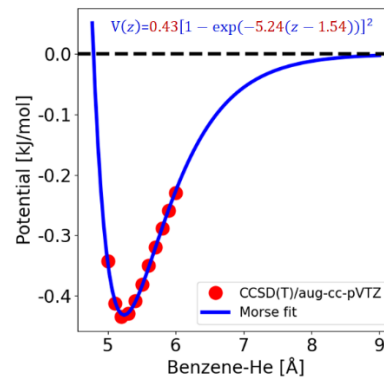
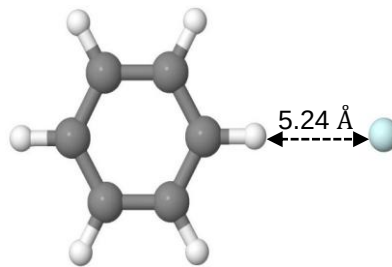
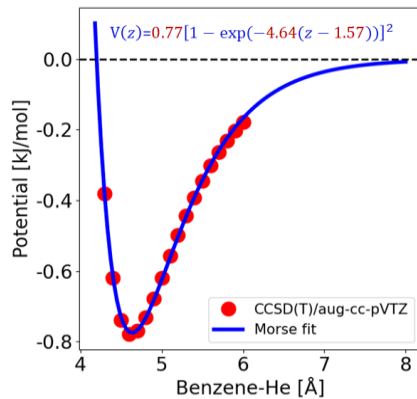
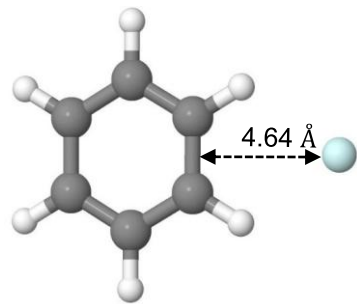
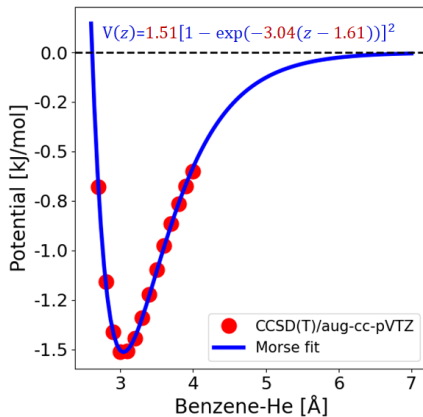
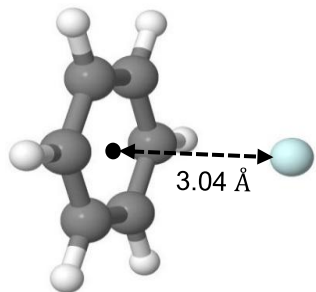
- | | |
|------------------------------|------------------------|
| ☞ Prof. Dr. Thomas Heine | ☞ MSc. Shabnam Haque |
| ☞ Prof. Dr. Knut R. Asmis | ☞ MSc. Florian Kreuter |
| ☞ Prof. Dr. Ralf Tonner-Zech | ☞ Dr. Lyuben Zhechkov |
| ☞ Prof. Dr. Daniel Boese | ☞ Dr. Knut Vietze |
| ☞ Dr. Toshiki Wulf | ☞ Mrs. Katrin Gerstner |
| ☞ Dr. Jiaye Jin | |

THANK YOU!

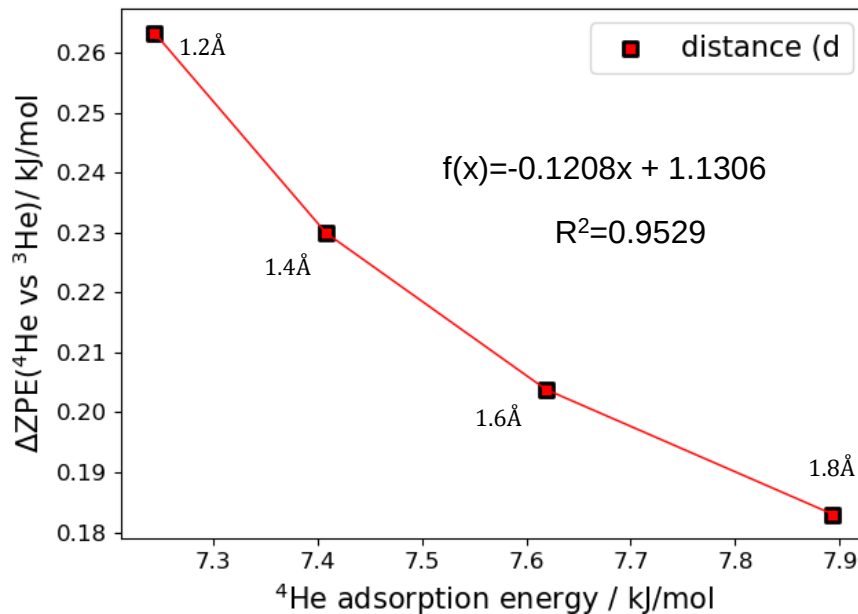
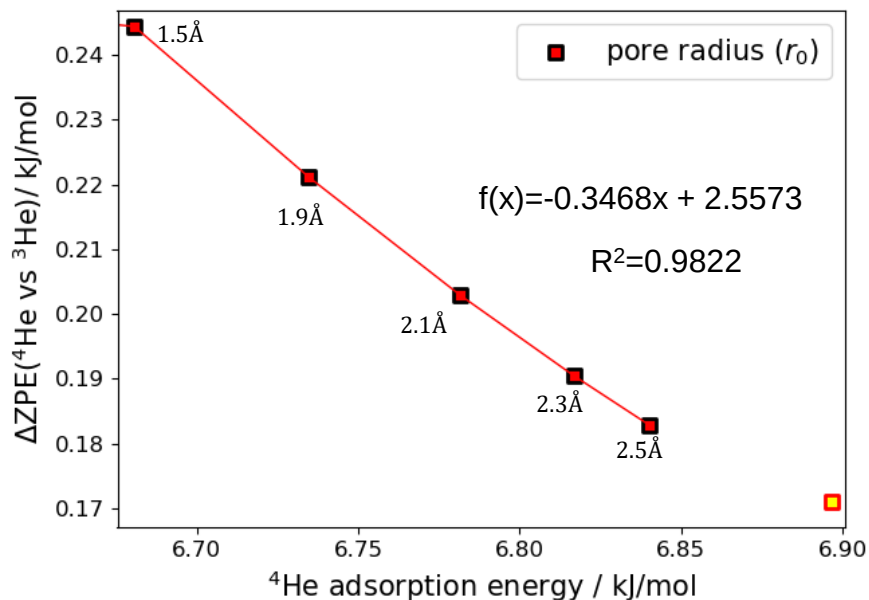


backup slides





Effect of pore size radius on ΔZPE and adsorption energy



$\Delta ZPE(^4\text{He}/^3\text{He}) \uparrow$ ΔZPE enhanced

Trade-off: $\Delta_{\text{ads}} E \downarrow$ binding weakened

1D and 3D Finite Difference Method (FDM)

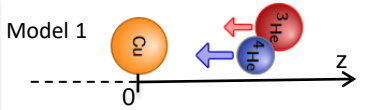
Solving Schrödinger-Equation

$$\hat{H}\Psi = E\Psi$$

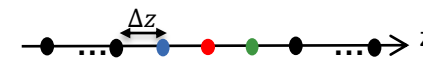
?

Via finite difference method

Model 1

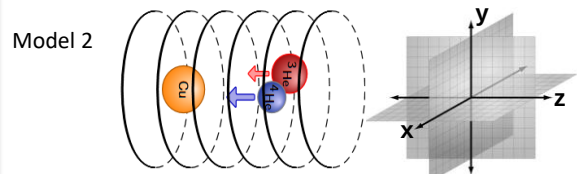


discretization

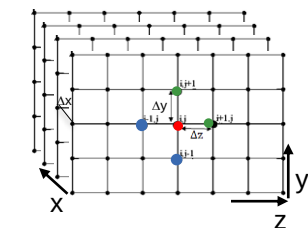


❖ **1D** $\hat{H} = -\frac{\hbar^2}{2m^*}\nabla^2 + V$ with $\nabla^2 = \frac{d^2}{dz^2}$

Model 2



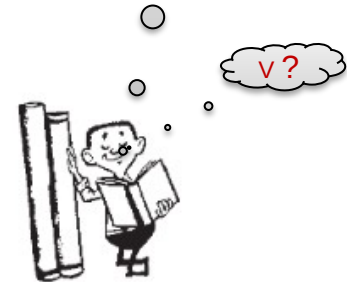
discretization



❖ **3D** $\hat{H} = -\frac{\hbar^2}{2m^*}\nabla^2 + V$ with $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

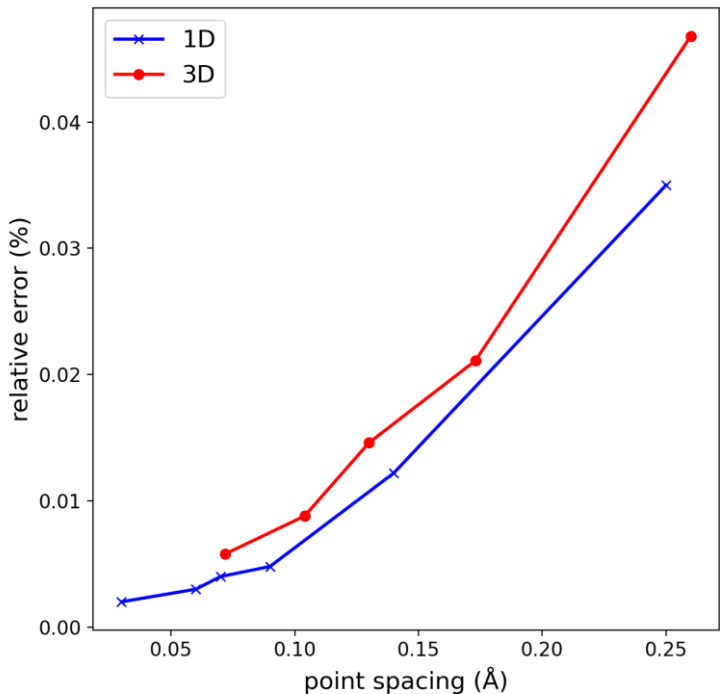
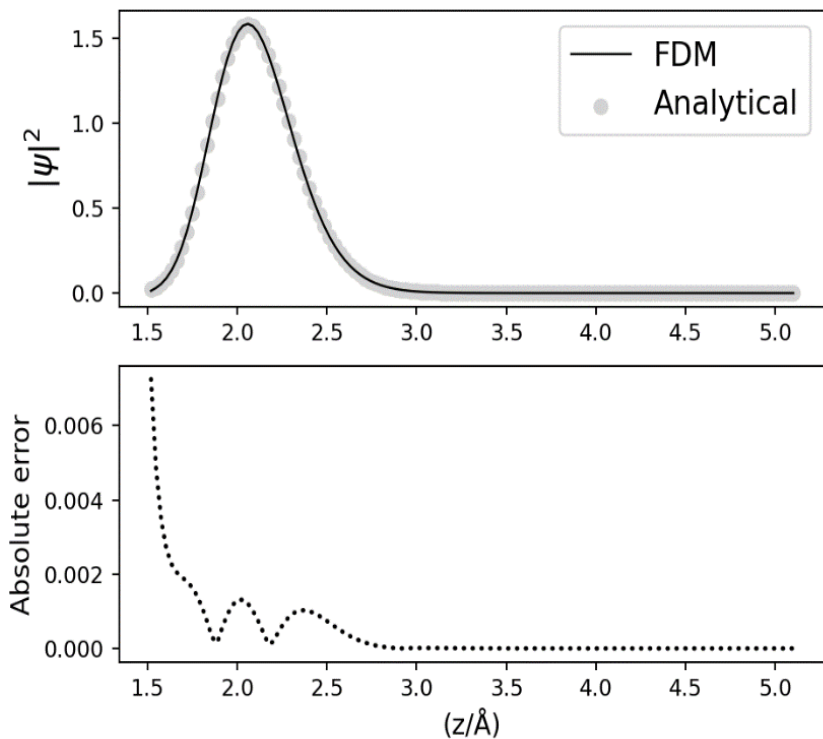
m^* : reduced mass; $\hbar$: reduced Planck constant

$$\hat{H} = -\frac{\hbar^2}{2m^*}\nabla^2 + V$$



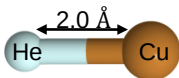
Computational approaches: Validation

$$\frac{E_{ZPE,anal} - E_{ZPE,num}}{E_{ZPE,anal}} \quad \text{1D}$$

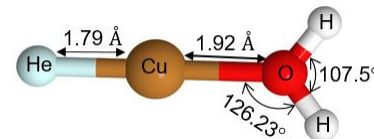


Computational approaches: Validation

Morse potential



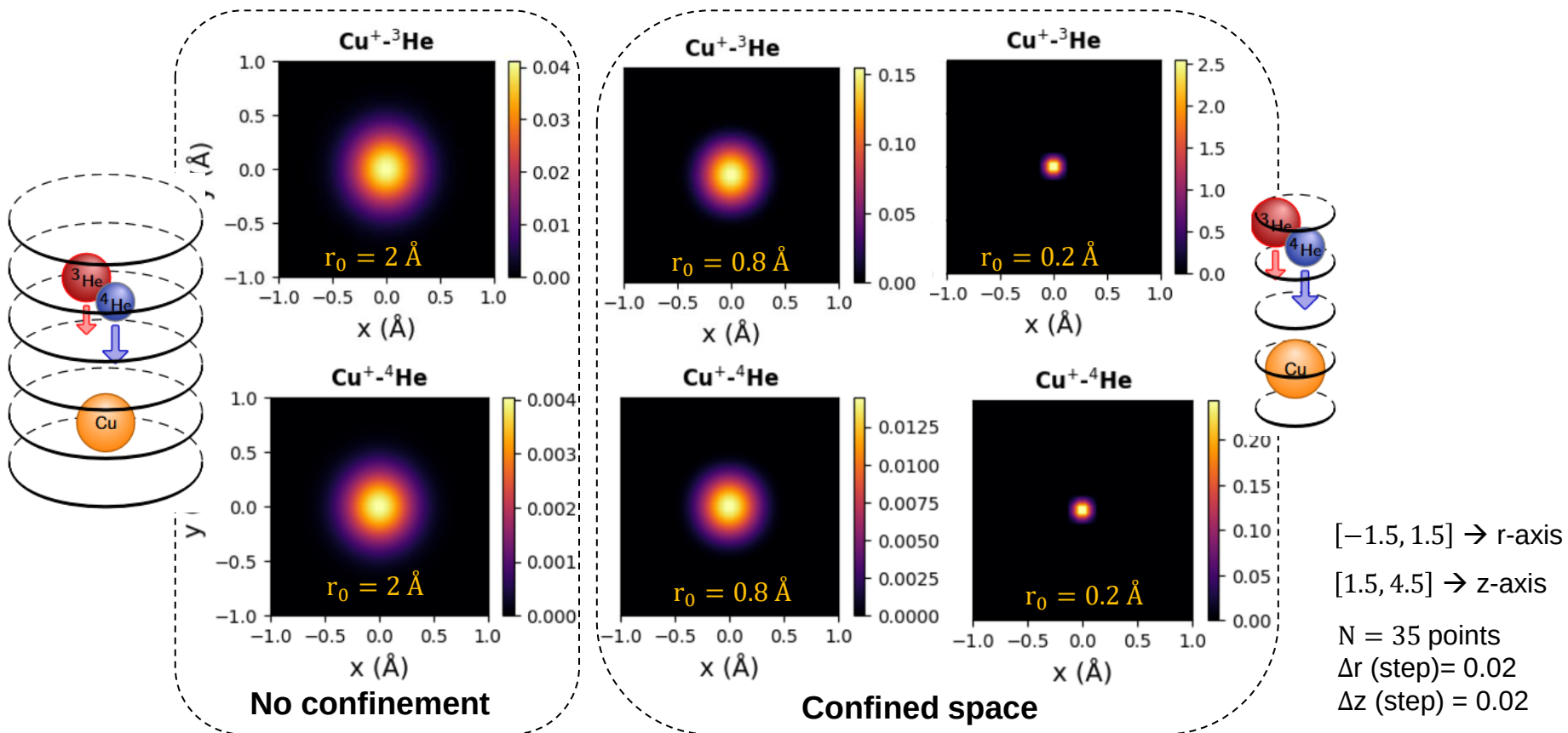
ZPE	Analytical solutions (kJ/mol)	Numerical solutions (kJ/mol)
		3D-FDM
⁴ He	1.296	1.293 ± 0.009
³ He	1.467	1.466 ± 0.009
Δ ZPE	0.171	0.166 ± 0.009



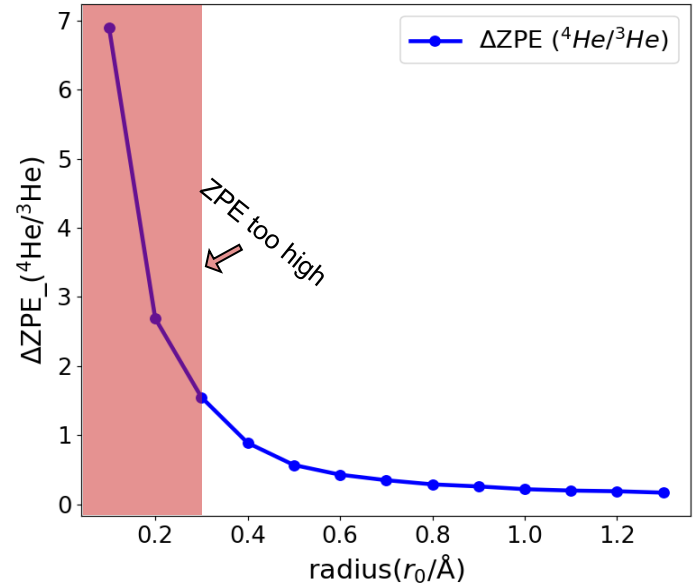
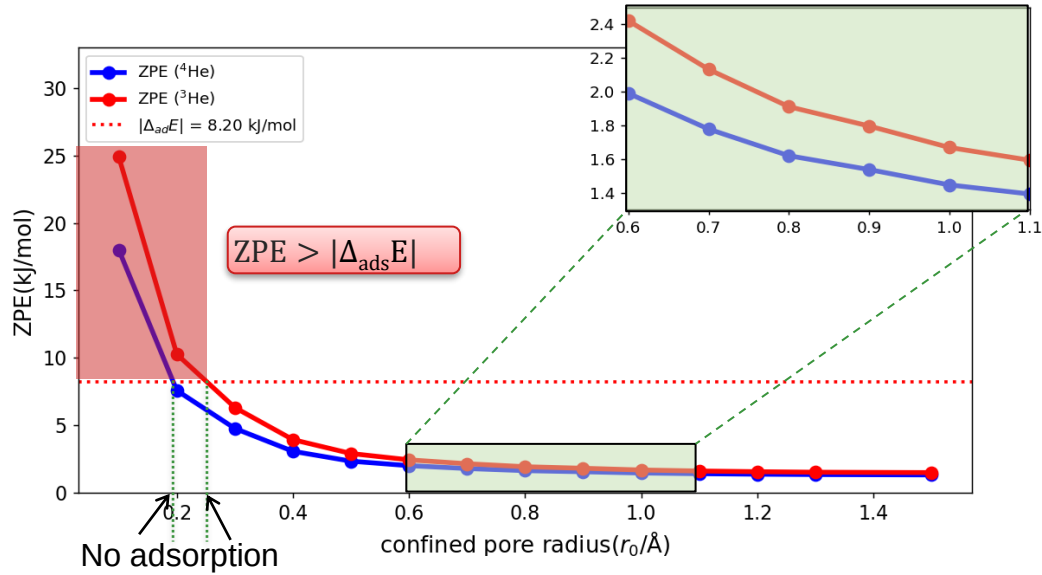
ZPE	Analytical solutions (kJ/mol)	Numerical solutions (kJ/mol)
		3D-FDM
⁴ He		± 0.009
³ He		± 0.009
Δ ZPE	0.171	± 0.009

0.009 is the truncation error $N = 66 \times 66 \times 66$ points; grid spacing: $\left\{ \begin{array}{l} \Delta r \text{ (step)} = 0.09 \text{ \AA} \\ \Delta z \text{ (step)} = 0.02 \text{ \AA} \end{array} \right.$

Effect of pore size radius on the probability density



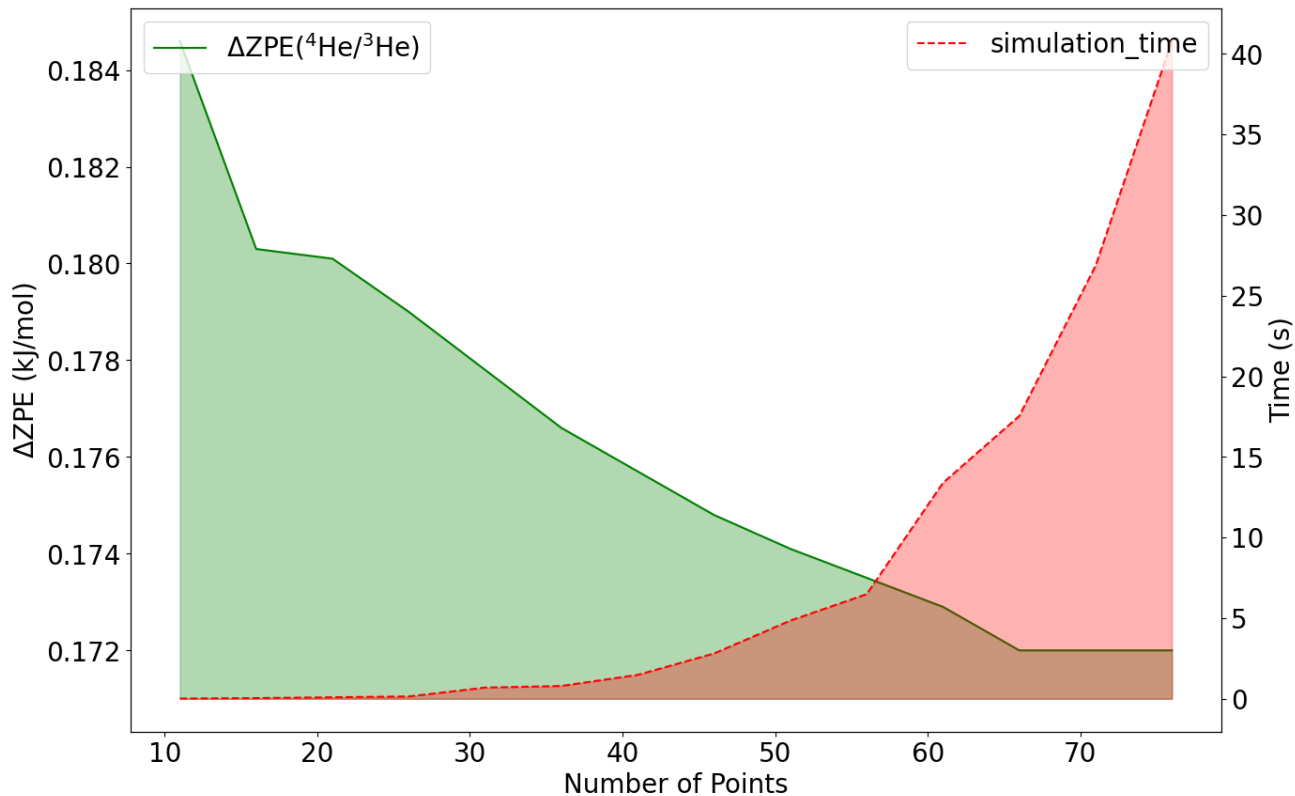
Effect of pore size radius on ΔZPE and adsorption energy



☞ Pore confinement can enhance ΔZPE

To what extend?

Computational cost: 3D-TISE with the Morse potential



3D-TISE: Ground state wave function

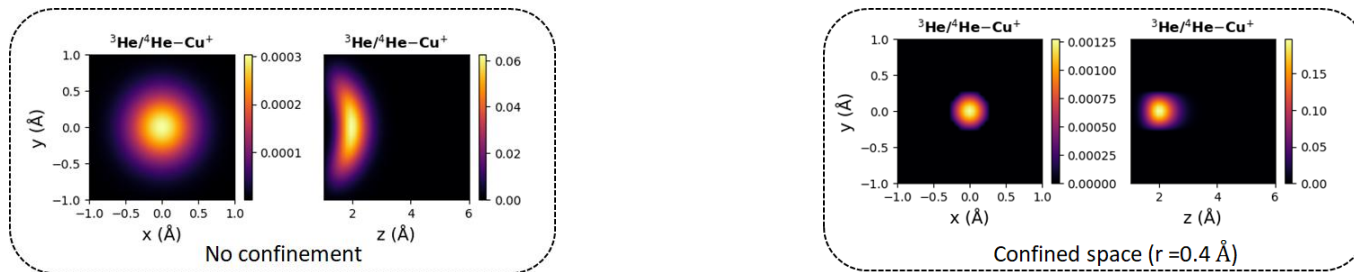
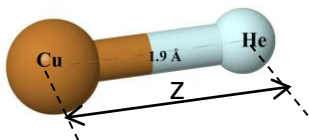


Figure 6: Density maps of the difference between the ground state wavefunction for He isotope bound to Cu(I). **Left:** without any confinement. **Right:** in presence of a pore confinement with the radius $r_0 = 0.3 \text{ Å}$. The presence of a confining pore potential reduces the displacement of the helium isotopes by increasing the probability of finding them near the nucleus. As we go closer the nuclei, we get a higher probability density

Interaction potential energy



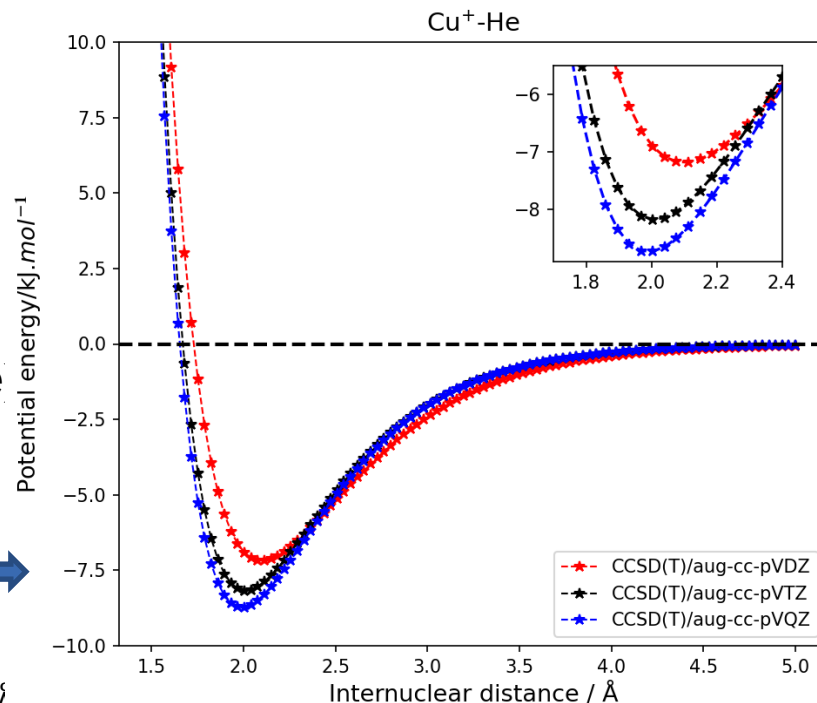
❖ Good approximation: **Morse potential**

$$V(z) = D_e \left[\left(1 - \exp(-\alpha(z - z_e)) \right)^2 \right]$$

D_e : dissociation energy, z_e : bond length, α (curvature)

Fitted Morse potential at the
CCSD(T)/aug-cc-pVTZ level of theory

$$D_e = 8.17 \text{ kJ}\cdot\text{mol}^{-1} \quad \alpha = 2.04 \text{ \AA}^{-1} \quad z_e = 2.011 \text{ \AA}$$



Crown complexes for He adsorption



Systems with Cu ⁺	Cu ⁺ ·He	(6c2) Cu ⁺ ·He	(9c3) Cu ⁺ ·He
Adsorption energy	−17.67	−12.78	−9.21
Systems with Ag ⁺	Ag ⁺ ·He	(6c2) Ag ⁺ ·He	(9c3) Ag ⁺ ·He
Adsorption energy	−5.78	−3.67	−0.56
Systems with Au ⁺	Au ⁺ ·He	(6c2) Au ⁺ ·He	(9c3) Au ⁺ ·He
Adsorption energy	−7.46	−6.05	−3.90
Systems with Li ⁺	Li ⁺ ·He	(6c2) Li ⁺ ·He	(9c3) Li ⁺ ·He
Adsorption energy	−9.40	− 6.28	− 4.37
Systems with Rh and Ir	/	C ₁₁ H ₈ NRh·He	C ₁₁ H ₈ NIr·He
Adsorption energy	/	− 1.75	− 104.70



Geometry: BLYP-D3(BJ)/TZP
Electronic energy : BLYP-D3(BJ)/TZP

Analytical solution

Morse potential : $V(z)=D_e \left[\left(1 - e^{(-a(z-z_e))} \right)^2 - 1 \right]$

Energy levels : $E_n = h\nu_n \left(n + \frac{1}{2} \right) - \frac{[h\nu_n (n + \frac{1}{2})]^2}{4D_e}$

and ZPE : $E_0 = \frac{h\nu_0}{2} - \frac{1}{4D_e} \left(\frac{h\nu_0}{2} \right)^2$

where:
$$\begin{cases} \nu_0 = \frac{a}{2\pi} \sqrt{2D_e/\mu} \\ n = 0, 1, 2 \dots \left[\lambda - \frac{1}{2} \right] \\ \lambda = \frac{\sqrt{2\mu D_e}}{a\hbar} \end{cases}$$

n : vibrational quantum number

ν : state with a unit of frequency

D_e : dissociation energy

μ : reduced mass; $\mu = \frac{m_1 m_2}{m_1 + m_2}$
 $(m_1 = m_{Cu}; m_2 = m_{He} \text{ or } m_{He})$

$\hbar$: reduced Planck constant $\left(\frac{h}{2\pi} \right)$